AI-POWERED INTELLIGENT TALENT ACQUISITION SYSTEM

INDIVIDUAL THESIS

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HIRING ATTRITION PREDICTION AND DECISION SUPPORT

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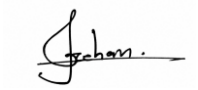
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ABSTRACT

Employee attrition creates significant operational and economic challenges for organizations and is an important consideration during recruitment because candidate suitability does not necessarily indicate long-term retention. This report presents the Hiring Attrition Prediction and Decision Support module of an AI-Powered Intelligent Talent Acquisition System, designed as an independent retention-risk layer operating alongside CV-to-job matching. The implemented solution uses a containerized microservices architecture comprising a Node.js/Express orchestrator, FastAPI-based machine learning services, a production CatBoost classifier, a secondary Logistic Regression model, TreeSHAP-based explainability, and a Gemini contextual explanation service. Candidate data are transformed into model-ready features through validation, normalization, and missing-value handling, while seven domain interaction ratios are engineered for the 52-feature CatBoost model. Two data lines are used: an adapted/synthetic dataset of 1,470 candidates for production CatBoost experiments and an empirical Sri Lankan workforce survey containing 835 responses. The production CatBoost model achieved a ROC-AUC of 0.8071 using five-fold stratified cross-validation at a tuned threshold of 0.33, while the empirical Sri Lankan Logistic Regression model achieved a ROC-AUC of 0.9029 and an F1-score of 64.15% on an untouched 167-response holdout set. The system further provides interactive What-If simulation for salary, work model, role/scope, and manager conditions, enabling recruiters to examine changes in predicted risk. The individual contribution includes dataset preparation, feature engineering, model development and evaluation, threshold selection, backend integration, explainability, What-If simulation, testing, and recruiter-facing UI development. The module provides explainable AI-based decision support while retaining the final recruitment decision with the human recruiter.

Keywords: Employee Attrition, Hiring Risk Prediction, CatBoost, Explainable AI, What-If Simulation

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
ATS	Applicant Tracking System
CV	Curriculum Vitae
F1	F1 Score
GBDT	Gradient Boosted Decision Trees
HR	Human Resources
LKR	Sri Lankan Rupee
MCC	Matthews Correlation Coefficient
ML	Machine Learning
OOF	Out-of-Fold
PR-AUC	Area Under the Precision-Recall Curve
ROC-AUC	Area Under the Receiver Operating Characteristic Curve
SaaS	Software as a Service
SHAP	SHapley Additive exPlanations
TA	Talent Acquisition
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

Recruitment has largely been based on evaluating whether the resume of the candidate is consistent with the job requirements. However, the consistency with a job and the likelihood of the candidate to stay in the organization are two different dimensions in recruitment. Even if a candidate possesses all the required qualifications, he or she might be at risk of leaving due to certain issues like remuneration, work environment, career perspectives, relationship with the managers, or other factors related to the workplace. In the context of Sri Lanka, predictors of turnover include factors such as job satisfaction, organizational support, job stress, alternatives, and overall commitment.

The audit of the repository shows that the component makes use of a polyglot containerized microservices architecture. Specifically, the frontend built in React and Vite sends the requests for predictions through an Express API Gateway to the Attrition Orchestrator. The orchestrator interacts with services of a production CatBoost model, an early-attrition Logistic Regression model, and an explainability service. Under the current configuration, the attrition capability is decoupled from the matching capability while making use of its output for predictions.

1.2 Individual component context

The Hiring Attrition solution is designed to address the issue of the second dimension of recruiting decisions. The solution does not affect the CV match score but gives extra attrition probability estimation with explanations of why a recruiter can think twice before giving an offer. The What-If simulation provides an addition to the prediction pipeline to allow the recruiter to play with the chosen offer and job conditions and see how it influences the prediction outcome.

My personal contributions to the project involved the following tasks: research statement, data preprocessing, feature engineering, modeling and benchmarking, thresholds tuning, orchestration on the backend, explainability, What-If simulation, testing, and integration of the recruiter's interface.

1.3 Problem statement

What would be an approach that allows this platform to use data collected from candidates, jobs, and labor market conditions to predict the probability of candidate dropout taking into consideration the class imbalance problem and incomplete applications, and still give the recruiter some feedback about risks and allow him/her to experiment with retention policies?

1.4 Research gap

Typically, recruitment qualification and retention risk are seen as two separate themes, which is evident in the platform designed as two distinct processes of CV matching and attrition prediction.The attrition research mainly focuses on the current employees, and here we consider the data on candidates and job offers to support the decision process in applications and offers evaluation.

The attrition study deals with the existing staff, while our focus lies on the data about candidates and job offers for the evaluation of the decisions made about applying and offering a candidate the job.

The models create a passive risk score, and in the project we conduct What-If analysis of the effect of the intervention on the model output.

There is also a lack of local research problem related to the lack of Sri Lankan recruitment datasets which include salary information, workplace, interview results and turnover outcomes for conducting machine learning studies. To solve this problem, this project utilizes the empirical Sri Lankan survey and generic benchmark datasets.

1.5 Aim and objectives

Design, build, and test an explainable machine-learning based decision support system which predicts the probability of attrition during different phases of the recruitment process, identifies significant risk factors, and allows recruiters to conduct what-if simulations based on pre-defined retention scenarios before making a human decision.

- Gather, clean and analyze empirical Sri Lankan workforce survey data related to employee turnover and workforce experience.
- Develop an adapter which maps candidates' data into the input space that has been used for training the models.
- Implement domain-specific interaction features which take into account equal pay, satisfaction, interviewing abilities, role tenure, manager tenure, and promotion timing.
- Train and benchmark CatBoost, Logistic Regression, XGBoost, LightGBM, and Random Forest models in the case of class imbalance.
- Implement decision thresholds derived from the training or cross-validation data instead of using artificial 0.50 values.
- Conduct What-if analysis for salary increase, work model changes, role/scope changes, and manager changes without modifying the candidate's database entry.
- Develop the prediction module and incorporate it into application analysis workflow for recruiters and test the behavior.

Remark: Modeling objectives and system objectives are purposefully different. Modeling objectives tell us whether the prediction is possible with the data available, while system objectives tell us whether the prediction can be communicated to the recruiter.

1.6 Research questions or hypotheses

How would such a system predict the probability of candidate drop out through analysis of data from the candidate, the position, and the labor market in light of difficulties with class imbalance and incomplete application processes, but at the same time enable the recruiter to determine the risk factors and test their retention methods without making an automatic hire?

1.7 Scope limitations and assumptions

This component deals with the estimation of attrition risk during the hiring process based on information about the candidate, job, interviewing process, salary, work environment, and other relevant sources. Production CatBoost line uses an adapted/synthetic benchmark, while local empirical line uses an 835-responses survey of Sri Lankan labor force. Each of these data sets has its own goals and is considered non-interchangeable evidence.

- Model output provides recommendations rather than makes the hiring decision itself.
- Missing candidate information might be required to impute; input coverage and imputed features are made visible in the system.
- What-If simulations cover only the intervention variables which are used in the system and cannot ensure the behavior in real world.
- Prospective validation based on longitudinal employer results is needed before making any claims.

1.8 Individual contribution

The following list represents my individual contributions to the project: problem statement, dataset preparation, feature extraction, modeling and benchmarking, threshold selection, backend orchestration, explainability, What-If simulation, testing, and recruiter interface integration.

1.9 Thesis structure

This thesis is organized according to the SLIIT individual thesis structure.

Chapter 1 introduces the platform context, individual component, problem, research gap, objectives, scope, and contribution.

Chapter 2 reviews the domain, related research, techniques, and conceptual gap.

Chapter 3 presents the research methodology, datasets, experimental procedure, evaluation measures, ethics, and reproducibility.

Chapter 4 describes product and solution design.

Chapter 5 explains implementation, integration, verification, and operation.

Chapter 6 presents research and product results.

Chapter 7 discusses the findings, contributions, limitations, and implications.

Chapter 8 considers commercialization and sustainability.

Chapter 9 concludes the work and identifies future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

This literature and related work review covers topics related to predicting employee turnover and attrition; Sri Lankan evidence on turnover intentions; machine-learning algorithms for structured human-resource data; explainable artificial intelligence; class imbalance assessment; threshold selection; and decision support during recruitment process. This review covers both technical approaches used in the component and the distinction between recruitment suitability and retention risk.

2.2 Domain and theoretical foundations

The above-mentioned facts create the background to develop a decision support tool in recruitment stage. In particular, the project is based on the idea of representing attrition as a probability, rather than a deterministic problem. Candidate's suitability is not equal to his/her retention risk. The outputs of the model are supplemented by explanations and what-if simulations which will allow recruiters to investigate possible intervention scenarios.

Typical human resource analytics tasks involve analysis of historical records of employees. While it allows identifying employees who are already integrated into an organization, the recruitment happens prior to acquiring longitudinal data about candidates. Thus, the proposed module takes into account the problem considering the data available at the recruitment stage, which includes salary expectation, work model, matching information, interview performance, job change, and some workplace factors.

The task of modeling is additionally complicated by the presence of class imbalance. Typically, in an attrition dataset, the size of the non-attrition class significantly outweighs the size of the positive class. Thus, a model optimized in terms of overall accuracy may seem good while neglecting quite a number of attrition candidates. This project, therefore, analyzes balanced accuracy, precision, recall, F1-score, ROC-AUC, and PR-AUC, and performs separate threshold tuning from probability estimation.

The gradient-boosted decision tree models show good results in dealing with structured human-resource data due to their capability to catch the non-linear dependencies and interaction between features. The CatBoost algorithm was chosen as the production model due to its capability to perform gradient boosting on categorical variables and to provide SHAP-compatibile explanations [1], [2]. The other algorithms used for benchmarking were Logistic Regression, XGBoost, LightGBM, and Random Forest [9], [10].

Hyperparameter optimization is performed using Optuna framework. The purpose is not just to tune the good configuration of a model but also to understand the behavior of a model under the imbalanced objective used in this research. The production configuration of the CatBoost model is tuned using PR-AUC metric with minimum recall constraint, while the empirical survey in Sri Lanka uses the approach of repeated cross-validation and training-only threshold tuning.

The What-If tool is another contribution of the project. Rather than stopping at the probability, the system allows adjusting the salary, work model, role/scope, and manager-related features. Each modification is processed using feature adaptation and inference, and thus, the simulation result is the reaction of the model to the modified input.

2.3 Related research

The issue of employee turnover has always been one of the organizational concerns. Sri Lankan researches show that there are several determinants which affect turnover intentions, such as job satisfaction, organizational support, job stress, alternative job opportunity availability, organizational commitment, and working environment. For example, a study of 228 IT-sector employees in Sri Lanka applies machine learning and explainable AI to analyze the turnover intention, whereas other Sri Lankan studies concern software engineer turnover, job satisfaction, work demands, and organizational factors [5]-[8].

The Sri Lankan evidence reviewed in the current thesis is the researches conducted on turnover intentions among startup professionals, software engineers, software developers, and employees in an apparel organization [5]-[8]. These studies motivate the consideration of job satisfaction, organizational support, stress, alternative opportunities, work demands, and other workplace factors.

2.4 Existing systems products and techniques

Typically, the standard human-resource analytics analyzes employees after they have joined an organization, whereas this component tackles the recruitment stage when longitudinal data is still not available. Machine-learning techniques analyzed in the project include Logistic Regression, XGBoost, LightGBM, Random Forest, and CatBoost. The system additionally combines local model attribution, contextual explanation, and scenario-based analysis, rather than being limited to a passive risk score.

2.5 Gap synthesis and conceptual framework

An additional local research problem is the absence of Sri Lankan recruitment datasets which contain salary, work environment, interview data, and turnover-related outcomes and which can be used for machine learning experiments. In order

to bridge this gap, this project uses the empirical Sri Lankan survey model together with a more generic benchmark dataset.

The conceptual gap bridged by the component is the separation of recruitment suitability from retention risk, together with a need for explanations and intervention-oriented scenario analysis. The solution combines the feature-adaptation layer, probability-based model inference, thresholded risk interpretation, feature-level evidence, contextual explanation, and What-If simulation within a human-in-the-loop recruiter workflow.

3 RESEARCH METHODOLOGY

3.1 Research design

The method is designed in accordance with the pipeline including candidate intake, feature preparation, model inference, explanation, and review by recruiters. The major decision made in terms of creating the method was connected with splitting of lines of the data set used in the model development process. In the case of CatBoost model, the benchmark will be the synthetic/adjusted benchmark data set consisting of 1,470 instances and 53 attributes, while for the empirical Sri Lankan study there will be a survey of 835 instances and 26 predictors.

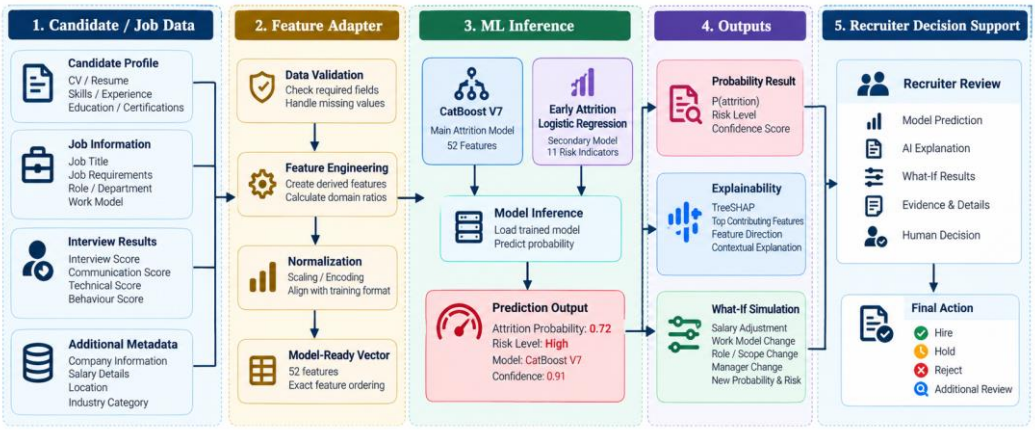


Figure 3.1. Candidate-to-Prediction Processing Pipeline

The processing flowchart is illustrated in Figure 3.1. At first, candidate and job data normalization is performed by the Feature Adapter that will handle missing values and scenario overrides as well as prepare the vector for the model. Model services will provide probability and risk signals while the explainability layer will provide feature level evidence. When recruiter changes a scenario, the same validated processing flowchart will be used and the result will be compared with the baseline.

3.2 Data participants or study materials

Two lines of data sets are provided. The empirical survey on Sri Lankan workforce is represented by 835 observations and 29 initial features. After deleting administrative features, 26 predicting features were used. The target ratio consists of 706 non-turnover observations (84.55%) and 129 turnover observations (15.45%) what provides imbalance ratio of 5.47:1. The data set includes Sri Lankan employment districts, salary feature in LKR and local notice period.

The second line of data set is the modified/adapted benchmark that consists of 1,470 observations and 53 features. It has standard structure of the IBM HR Analytics

attrition benchmark with regional hiring features, such as salaries in LKR, interviews and workplace policies. The target has 1,233 negative observations and 237 positive ones (16.12% positive ratio). It is used for CatBoost model and its ablations production and can't be considered the data set of Sri Lankan employee exits confirmation.

Table 3.2. Dataset comparison summary

Dataset line	Records	Predictors / columns	Positive class	Primary use
Empirical Sri Lankan survey	835	26 predictors / 29 original columns	129 turnover (15.45%)	Local empirical model and holdout evaluation
Adapted / synthetic benchmark	1,470	53 columns	237 attrition (16.12%)	CatBoost benchmark and feature ablation

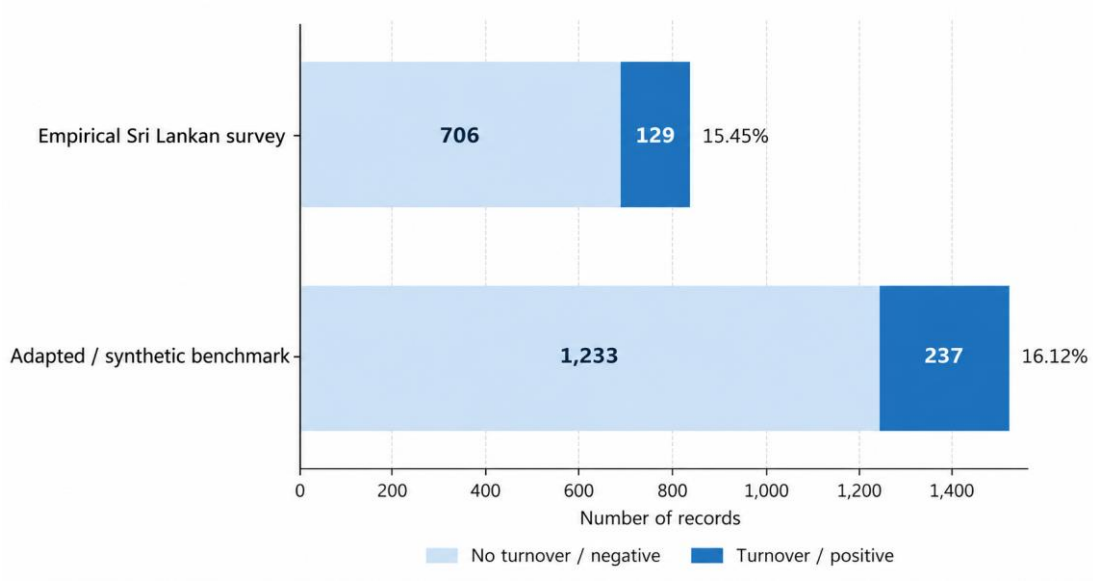


Figure 3.2. Dataset Lines Used in the Attrition Research

3.3 Experimental or study procedure

The production CatBoost model works with 52 features: 45 adapted or raw variables and seven engineered interaction ratios of the domain. The engineered ratios are computed after the candidate-field normalization and imputation in order to get a consistent view of compensation, satisfaction, interview, tenure, management stability, and career advancement.

Table 3.1. Model feature engineering specification

Feature	Definition	Purpose
OfferExpectedRatio	OfferedSalaryLKR / ExpectedSalaryLKR	Measures compensation parity
AverageSatisfaction	Mean of environment, job, relationship, and work-life satisfaction	Composite morale indicator
InterviewComponentAverage	Mean of technical, communication, behaviour, and confidence scores	Summarizes interview evidence
MatchInterviewAverage	(MatchScore + InterviewScore) / 2	Combines suitability and interview evidence
RoleTenureRatio	YearsInCurrentRole / (YearsAtCompany + 1)	Represents role stagnation
ManagerTenureRatio	YearsWithCurrManager / (YearsAtCompany + 1)	Represents management stability

The secondary Early Attrition service uses 11 normalized risk indicators: SalaryGapRisk, CompanySwitchRate, LowMatchRisk, LowSimilarityRisk, LowInterviewRisk, NoticeRisk, LocationWorkRisk, TrainingRisk, MentorshipRisk, CareerDevelopmentRisk, and CertificationRisk. These indicators are another source of early risk signal and are evaluated separately from the production CatBoost model.

For development, the project uses local development environment in addition to containerized runtime services. The production architecture design allows for separating testing of the model layer, orchestration layer, and UI.

For experiments, stratified validation was used for the imbalanced benchmark experiment, and holdout evaluation was used for the empirical Sri Lankan survey model. The hyperparameters and threshold were chosen using training or cross-validation data rather than choosing the 0.50 default threshold.

3.4 Measures and evaluation metrics

The evaluation is based on the five-fold stratified cross-validation in case of the benchmark experiment. As the benchmark experiment is imbalanced, the evaluation focuses on balanced accuracy, precision, recall, F1, ROC-AUC, and PR-AUC rather than only accuracy.

The reporting uses the accuracy, balanced accuracy, precision, recall, F1-score, ROC-AUC, PR-AUC, Matthews Correlation Coefficient (MCC), and specificity when appropriate. Thresholds performance is evaluated separately from continuous probabilities.

3.5 Data analysis

Empirical Sri Lankan survey model was trained from 835 survey responses using 80/20 stratified split. Repeated stratified five-fold cross-validation with two repetitions was used for model selection. Logistic Regression had the maximum PR-AUC in repeated-CV of 0.85227 among the four candidate architectures.

The project doesn't assume the standard 0.50 classification threshold for all models. For the production CatBoost benchmark, the threshold of 0.33 was selected under 0.50 minimum recall constraint. The CatBoost recall on the standard 0.50 cutoff is 35.46%; on 0.33 – 50.62%, and F1 increases from 45.50% to 50.34%. For the Sri Lankan survey Logistic Regression model, out-of-fold F1-based search selected the threshold of 0.23, producing 65.38% recall and 62.96% precision on the untouched holdout test.

3.6 Ethics privacy security and responsible AI

- The system labels the attrition output as a prediction/risk signal, not the deterministic statement about the candidate.
- The candidate match score is separated from the attrition risk.
- The recruiters are still responsible for the decision; the AI doesn't automate hiring or rejection of candidates.
- The demographic variables such as age, gender, and marital status were excluded from the primary empirical model feature set.
- The FeatureAdapter reports the input coverage and imputed features in order to avoid hidden uncertainty due to missing data.
- Model explanations are based on model attribution and candidate/job context, not allowing for the re-calculation of the prediction by the narrative model.

3.7 Reproducibility and research artefacts

The research artefacts are the prepared dataset lines, feature-adaptation logic, engineered-features specifications, model training and benchmark configurations,

threshold values, model files, inference services, and application integration. The benchmark and empirical model lines are kept separate in order to preserve the provenance and purpose of each result. The reporting of experiments and threshold choices is done alongside the software and deployment technologies used by the component.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

- Provide an authenticated recruiter-facing attrition prediction entry point.
- Transform candidate and job information into a stable model-ready representation while handling missing values.
- Return probability, risk score, risk level, threshold metadata, and contributing factors.
- Provide a secondary early-attrition signal without replacing the primary model.
- Provide grounded explanations based on model drivers and candidate/job context.
- Support What-If scenario analysis without modifying the stored candidate record.
- Keep candidate suitability/matching separate from retention-risk estimation.
- Maintain a human-in-the-loop final recruitment decision.

4.2 Component architecture within the complete solution

The design follows a polyglot microservices approach in which the recruiter-facing interface communicates through an API Gateway, the Attrition Orchestrator prepares stable prediction inputs, and the model services perform independent inference. This structure supports separation of concerns between interface, orchestration, and model development.

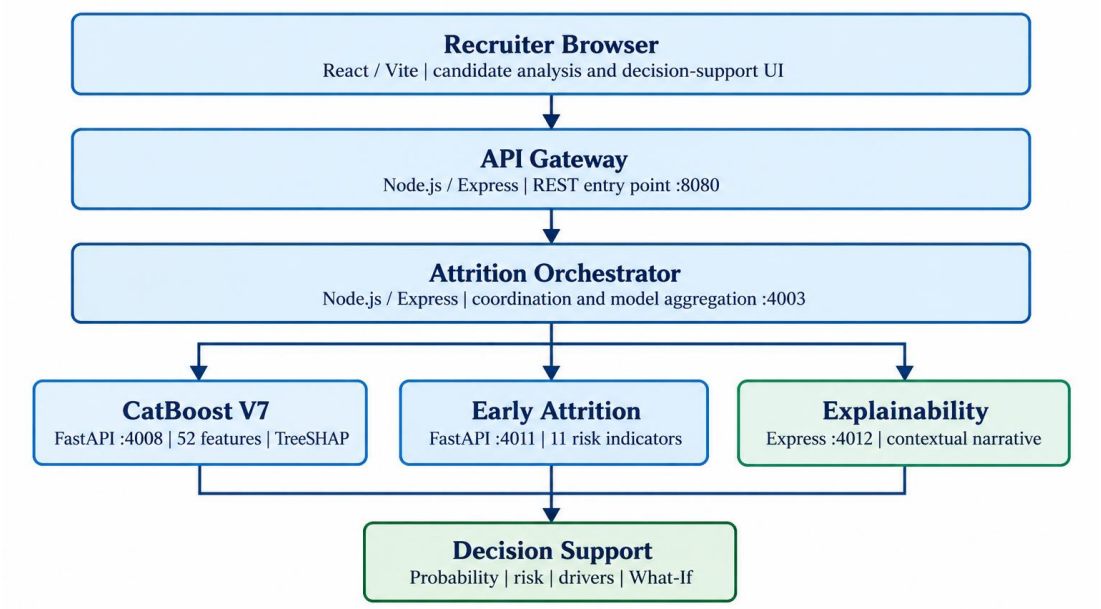


Figure 4.1. Hiring Attrition Decision-Support Architecture

Architecture design is intended to follow a polyglot microservices approach. The frontend uses React/Vite, the API Gateway uses Node.js and Express, and the Attrition Orchestrator coordinates the CatBoost, Early Attrition, and explainability services. Model services expose independent inference interfaces, and model artifacts are loaded independently of orchestration. This separation enables model development and validation to proceed independently from the recruiter interface and allows model artifacts to be updated without changing the interface.

4.3 Design alternatives and decisions

A separate attrition layer was selected rather than modifying the existing CV-to-job matching score. This preserves two distinct evidence dimensions: suitability for the role and likelihood of retention. CatBoost was selected as the operational production model because the research required structured-data performance, categorical-feature handling, and local explainability. Logistic Regression was retained as the empirical Sri Lankan model line. The design separates model attribution from contextual language generation so that the explanatory service does not become a second scoring system.

A polyglot microservices architecture was selected to isolate model inference, orchestration, and interface development. What-If simulation reuses the same validation and feature-adaptation path rather than implementing a separate calculation, and a rule-based fallback is retained as a service-continuity mechanism rather than being treated as an equivalent trained-model result.

4.4 Detailed design

Table 4.1. API endpoint specification

Endpoint	Service	Purpose
POST /api/attrition/predict	API Gateway	Authenticated external entry point
POST /predict	Attrition Orchestrator	Coordinates prediction services
POST /predict	CatBoost Model Service	52-feature CatBoost inference + TreeSHAP
POST /predict	Early Attrition Service	11-feature Logistic Regression inference
POST /explain-attrition	Explainability Service	Grounded explanation from model drivers + CV/job context
GET /health	All services	Health and model-load verification

The API Gateway exposes the external /api/attrition/predict entry point and applies gateway-level authentication and rate limiting. The Attrition Orchestrator validates the request, calculates early-attrition features, calls model services in parallel, combines their results, and invokes the explanation service when contextual explanation is required. The CatBoost service exposes POST /predict and returns probability, risk score, risk level, threshold, model identifier, and local drivers. The Early Attrition service returns its independent prediction and drivers. The explanation service exposes POST /explain-attrition.

The orchestration layer performs a second important function: it converts the application-level candidate representation into a stable contract understood by the prediction services. This reduces coupling between upstream product fields and the trained model schema. The same contract is reused during What-If simulation, which helps ensure that the baseline and scenario predictions are produced through comparable processing paths.

The recruiter-facing implementation contains WhatIfSimulation.jsx, RiskMeter.jsx, AttritionExplainability.jsx, and ModelResults.jsx. The What-If component manages the intervention controls and baseline/scenario comparison. RiskMeter renders the probability as a visual risk gauge. AttritionExplainability renders model summaries, thresholds, TreeSHAP contribution indicators, and contextual narrative. ModelResults provides the technical audit information required to inspect the target, model version, probability, threshold, and classification result.

The interface is intentionally structured around separate evidence dimensions. Match score, interview score, attrition probability, confidence, and career stability are displayed independently rather than collapsed into a single automatic hiring score. This supports a human-in-the-loop workflow and reduces the risk of interpreting a retention prediction as a direct judgement of candidate suitability.

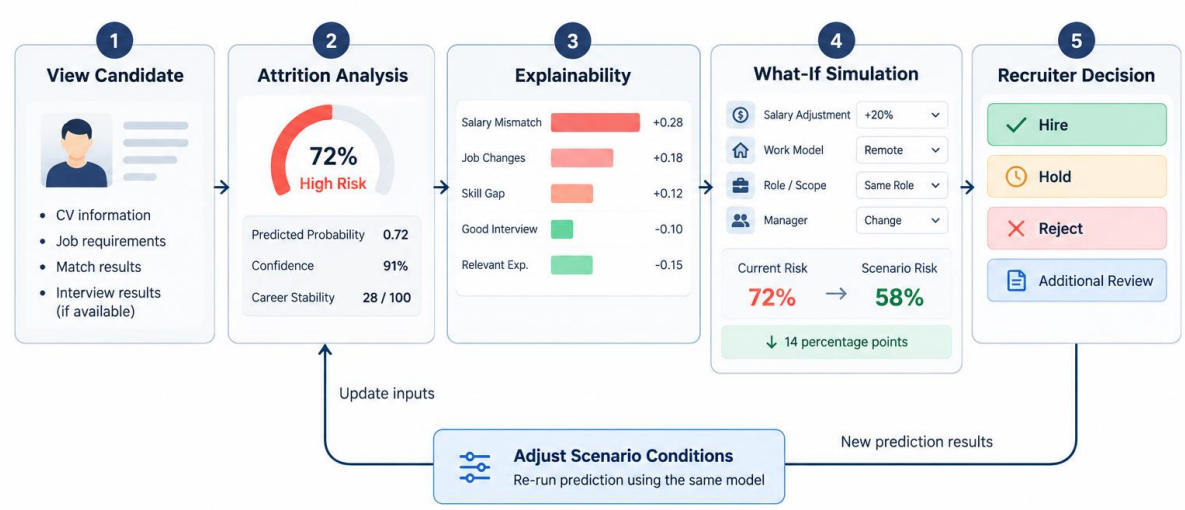


Figure 4.2. Recruiter User Workflow

4.5 Standards and professional practice

The solution follows professional practices of separation of concerns, modular service boundaries, explicit API contracts, validation before inference, reproducible evaluation, and human oversight of consequential decisions. Security-related practices include authenticated external access, separation of internal model services from the public gateway, controlled use of secrets, and explicit communication of model and threshold metadata. The system is designed to support inspection and review rather than conceal uncertainty behind a single automated hiring score.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

The project combines web application development, backend service orchestration, machine-learning model training, explainability, and containerized deployment. The implementation uses the local resources and runtime technologies summarized in Table 5.1.

Table 5.1. Materials and tools

Category	Technology / Resource	Purpose
Frontend	React 18, Vite, CSS Modules	Recruiter UI, risk meter, What-If simulation
API / Gateway	Node.js, Express	Gateway and orchestration
ML API	Python, FastAPI, Uvicorn	Model inference services
ML Libraries	CatBoost, scikit-learn, XGBoost, LightGBM	Model training and evaluation
Data Processing	Pandas, NumPy	Data cleaning, feature engineering, transformation
Explainability	TreeSHAP, model coefficients, Gemini explanation service	Local drivers and contextual explanation
Optimization	Optuna	Hyperparameter optimization
Persistence	Model artifacts, joblib, project data files	Model and metadata storage
Deployment	Docker / Docker Compose	Containerized service deployment

5.2 Component implementation

The API Gateway exposes the external /api/attrition/predict entry point and applies gateway-level authentication and rate limiting. The Attrition Orchestrator validates the request, calculates early-attrition features, calls model services in parallel, combines their results, and invokes the explanation service when contextual explanation is required. The CatBoost service exposes POST /predict and returns probability, risk score, risk level, threshold, model identifier, and local drivers. The Early Attrition service returns its independent prediction and drivers. The explanation service exposes POST /explain-attrition.

The orchestration layer performs a second important function: it converts the application-level candidate representation into a stable contract understood by the prediction services. This reduces coupling between upstream product fields and the trained model schema. The same contract is reused during What-If simulation, which

helps ensure that the baseline and scenario predictions are produced through comparable processing paths.

The recruiter-facing implementation contains WhatIfSimulation.jsx, RiskMeter.jsx, AttritionExplainability.jsx, and ModelResults.jsx. The What-If component manages the intervention controls and baseline/scenario comparison. RiskMeter renders the probability as a visual risk gauge. AttritionExplainability renders model summaries, thresholds, TreeSHAP contribution indicators, and contextual narrative. ModelResults provides the technical audit information required to inspect the target, model version, probability, threshold, and classification result.

The interface is intentionally structured around separate evidence dimensions. Match score, interview score, attrition probability, confidence, and career stability are displayed independently rather than collapsed into a single automatic hiring score. This supports a human-in-the-loop workflow and reduces the risk of interpreting a retention prediction as a direct judgement of candidate suitability.

5.3 Integration with the other components

The auditing of the repository shows that the component uses a polyglot, containerized microservices architecture. The frontend built with React and Vite sends prediction requests via an Express API Gateway to the Attrition Orchestrator. This orchestrator interacts with the services of a production CatBoost model, early-attrition Logistic Regression model, and an explainability service. This way, the attrition functionality stays independent from the matching one while using its output for predictions when available.

The FeatureAdapter maps and normalizes the candidate information, applies any current simulation state, imputes missing fields, and builds the model vector.

The CatBoost and Early Attrition services perform their predictions in parallel.

The unified result is returned to the frontend and displayed as probability, risk level, threshold status, model agreement, and contributing factors.

The recruiter can run What-If scenarios without mutating the stored candidate snapshot.

The final employment or offer decision is made by the recruiter rather than by the model.

5.4 Verification and software testing

Table 5.2. Functional test cases

Test Case	Expected Behaviour	Observed / Verified Outcome	Status
Valid candidate prediction	Return probability, risk score, level, threshold metadata	Prediction service response verified	Pass
Missing numerical field	Use training median and expose imputation	inputCoverage and imputedFeatures returned	Pass
Missing categorical field	Use explicit Missing category	Model vector remains valid	Pass
Salary What-If	Recalculate salary-derived inputs and re-predict	Scenario output updates dynamically	Pass
Role/Scope What-If	Apply career-development scenario	Scenario output re-enters prediction path	Pass
Manager What-If	Apply mentorship-related scenario	Scenario output updates dynamically	Pass
Remote work What-If	Change work model and re-predict	Scenario output updates dynamically	Pass
Explainability request	Return model drivers without changing score	SHAP drivers and grounded narrative returned	Pass
Model service unavailable	Use configured service-continuity fallback	Fallback path available in orchestrator	Pass
Health endpoint	Expose service/model readiness	GET /health used for verification	Pass

The functional test evidence reported in the existing thesis verifies prediction, missing-value handling, scenario updates, explainability, service-continuity fallback, and health/readiness behavior.

5.5 Deployment and operation

The application uses Docker and Docker Compose as the repeatable deployment mechanism. The production architecture is intentionally modular so that the model layer, orchestration layer, and user interface can be tested separately. Model artefacts are loaded independently of orchestration and can therefore be replaced without changing the recruiter interface.

Table 5.3. Software and deployment tools

Application UI	React 18, Vite, CSS Modules	Candidate analysis, risk visualization, What-If controls
Service Layer	Node.js, Express, Fetch	API Gateway and attrition orchestration
ML Services	Python, FastAPI, Uvicorn	Model loading and inference
Models	CatBoost, Logistic Regression	Primary and secondary prediction services
Data Science	Pandas, NumPy, scikit-learn	Preparation, transformation, evaluation
Explainability	TreeSHAP, coefficients, Gemini service	Local attribution and contextual narrative
Optimization	Optuna	Hyperparameter search
Deployment	Docker / Docker Compose	Container orchestration and repeatable deployment
Version Control	Git / GitHub	Source management and collaboration

6 RESULTS AND VALIDATION

6.1 Research results

Five-fold stratified cross-validation was used for the benchmark experiment. Because the benchmark is imbalanced, the evaluation emphasizes balanced accuracy, precision, recall, F1, ROC-AUC, and PR-AUC rather than accuracy alone.

Table 6.1. Production model benchmark results

Model	Threshold	Accuracy (%)	Balanced Acc. (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC	PR-AUC
Optuna CatBoost	0.33	83.81 ± 1.90	70.40 ± 1.50	50.44 ± 5.55	50.62 ± 2.72	50.34 ± 3.11	0.8071 ± 0.0239	0.5753 ± 0.0475
Optuna CatBoost	0.50	86.33 ± 1.48	65.78 ± 2.24	65.33 ± 11.14	35.46 ± 4.78	45.50 ± 4.77	0.8071 ± 0.0239	0.5753 ± 0.0475
Default CatBoost	0.50	86.67 ± 0.84	61.04 ± 2.61	81.74 ± 11.91	23.22 ± 5.67	35.58 ± 6.60	0.8218 ± 0.0244	0.5974 ± 0.0538
XGBoost	0.50	86.46 ± 0.59	64.15 ± 2.07	67.73 ± 4.36	31.21 ± 4.76	42.45 ± 4.14	0.8045 ± 0.0330	0.5385 ± 0.0514
LightGBM	0.50	86.33 ± 1.02	63.73 ± 1.57	67.28 ± 7.83	30.39 ± 2.92	41.74 ± 3.58	0.7981 ± 0.0381	0.5338 ± 0.0609
Random Forest	0.50	85.99 ± 0.95	57.39 ± 2.50	89.33 ± 13.73	15.19 ± 4.91	25.62 ± 7.37	0.8087 ± 0.0315	0.5443 ± 0.0844
ElasticNet Logistic	0.50	87.55 ± 0.59	63.95 ± 0.94	82.54 ± 6.75	29.12 ± 1.71	43.00 ± 2.28	0.8267 ± 0.0236	0.6163 ± 0.0326

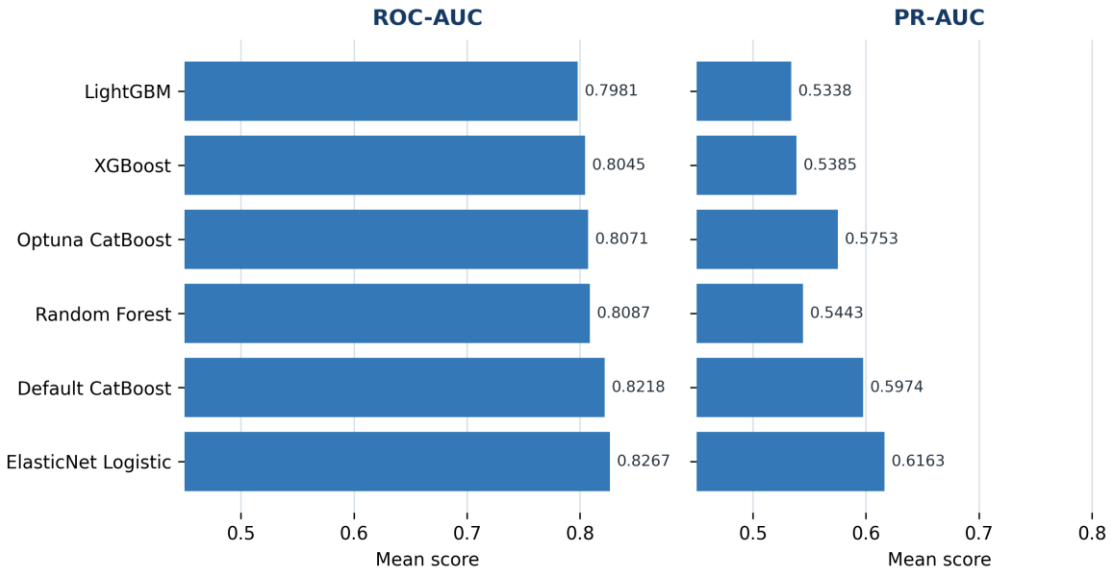


Figure 6.1. ROC-AUC and PR-AUC across five-fold benchmark configurations

The results show that the optimized CatBoost configuration at threshold 0.33 substantially improves recall relative to the standard 0.50 cutoff while sacrificing some raw accuracy. This illustrates why threshold selection matters in an imbalanced attrition problem. The benchmark table also shows that the highest ROC-AUC and PR-AUC were not obtained by the production CatBoost configuration, so the CatBoost model should not be described as universally superior to all baselines. The ROC-AUC and PR-AUC comparison is visualized in Fig. 5.

In practical terms, the benchmark supports two conclusions. First, the attrition task benefits from evaluation measures that account for minority-class detection. Second, model choice should consider operational fit and explainability in addition to a single ranking metric. The production configuration was therefore operationalized as a risk-scoring service while retaining the other algorithms as explicit research baselines.

Table 6.2. Feature ablation results

Configuration	Features	Accuracy (%)	Balanced Acc. (%)	Precision (%)	Recall (%)	F1 (%)	ROC-AUC	PR-AUC
Standard IBM	26	84.22 ± 0.90	73.21 ± 1.20	51.15 ± 2.44	56.96 ± 3.44	53.78 ± 1.26	0.8160 ± 0.0204	0.5834 ± 0.0402
Raw + Regional	45	85.58 ± 2.41	72.14 ± 1.39	58.07 ± 11.27	52.31 ± 4.00	54.18 ± 3.29	0.8142 ± 0.0086	0.5817 ± 0.0517

Full + 7 Ratios	52	83.81 ± 1.90	70.40 ± 1.50	50.44 ± 5.55	50.62 ± 2.72	50.34 ± 3.11	0.8071 ± 0.0239	0.5753 ± 0.0475
Delta B - A	+19	+1.36	-1.07	+6.91	-4.65	+0.40 (p=0.830)	-0.0018 (p=0.821)	-0.0017 (p=0.933)
Delta C - B	+7	-1.77	-1.74	-7.63	-1.69	-3.85 (p=0.026)*	-0.0071 (p=0.463)	-0.0063 (p=0.575)

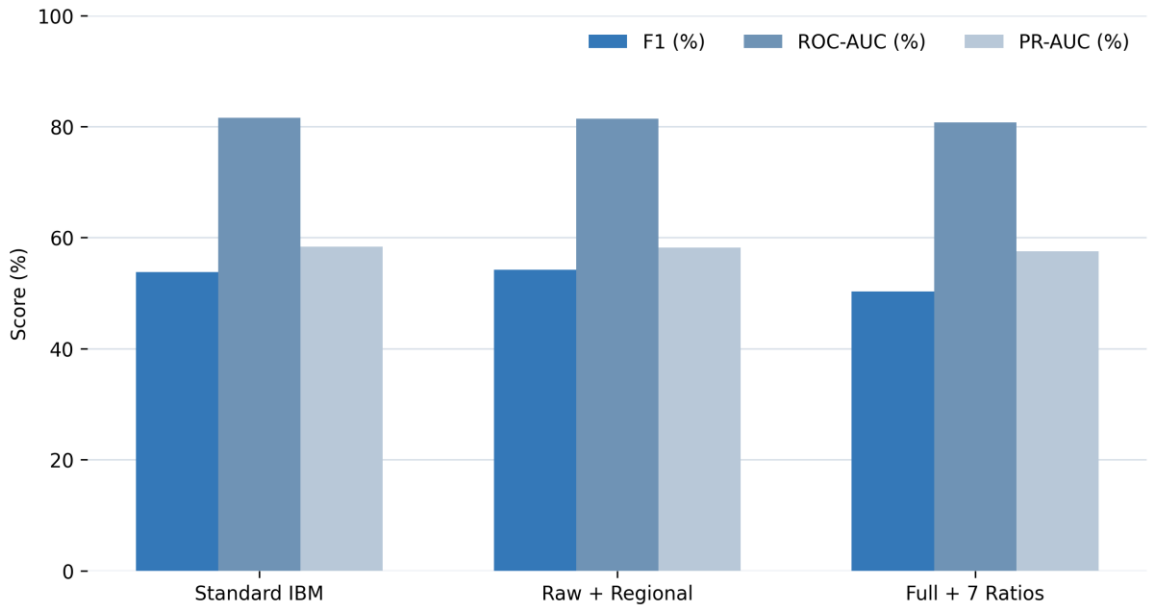


Figure 6.2. Feature Ablation Performance

Paired tests reported a +0.40% F1 change from Standard IBM to Raw + Regional (p=0.830) and a -3.85% F1 change from Raw + Regional to Full + 7 Ratios (p=0.026). The latter difference was statistically significant in the reported experiment.

The ablation result is methodologically important because it shows that adding engineered ratios did not automatically improve performance. Feature engineering

should therefore be validated empirically, especially under small or imbalanced datasets, rather than justified only by domain intuition.

The empirical Sri Lankan survey model was trained from 835 survey responses using an 80/20 stratified split. Repeated stratified five-fold cross-validation with two repeats was used for model selection. Logistic Regression produced the highest repeated-CV PR-AUC of 0.85227 among the four candidate architectures.

Table 6.3. Empirical Sri Lankan survey model results

Metric	Holdout Result
Test samples	167
Accuracy	88.62%
Precision	62.96%
Recall	65.38%
F1-Score	64.15%
Balanced Accuracy	79.15%
ROC-AUC	0.9029
PR-AUC	0.6577
MCC	0.5741
Specificity	92.91%
Decision Threshold	0.2300

These results provide the strongest direct empirical evidence in the repository for a Sri Lankan attrition-related model. They should be reported as results of the survey model rather than as performance of the production CatBoost service, because the two models use different datasets, features, and training protocols.

Table 6.4. Secondary early attrition model results

Metric	Result
Accuracy	85.71%
Precision	77.98%
Recall	82.52%
F1-Score	80.19%
ROC-AUC	0.9045
PR-AUC	0.8448
MCC	0.6910
Decision Threshold	0.3930

The secondary Early Attrition artifact produces a strong recall-oriented result on its test set. Its purpose in the application is to provide an additional risk signal based on 11 normalized indicators rather than to replace the primary CatBoost model.

The project does not rely on the standard 0.50 classification threshold for every model. For the production CatBoost benchmark, a threshold of 0.33 was selected under a minimum recall constraint of 0.50. At the standard 0.50 cutoff, CatBoost recall was 35.46%; at 0.33, recall increased to 50.62% and F1 increased from 45.50% to 50.34%. For the Sri Lankan survey Logistic Regression model, an out-of-fold F1 search selected a threshold of 0.23, which produced 65.38% recall and 62.96% precision on the untouched holdout test.

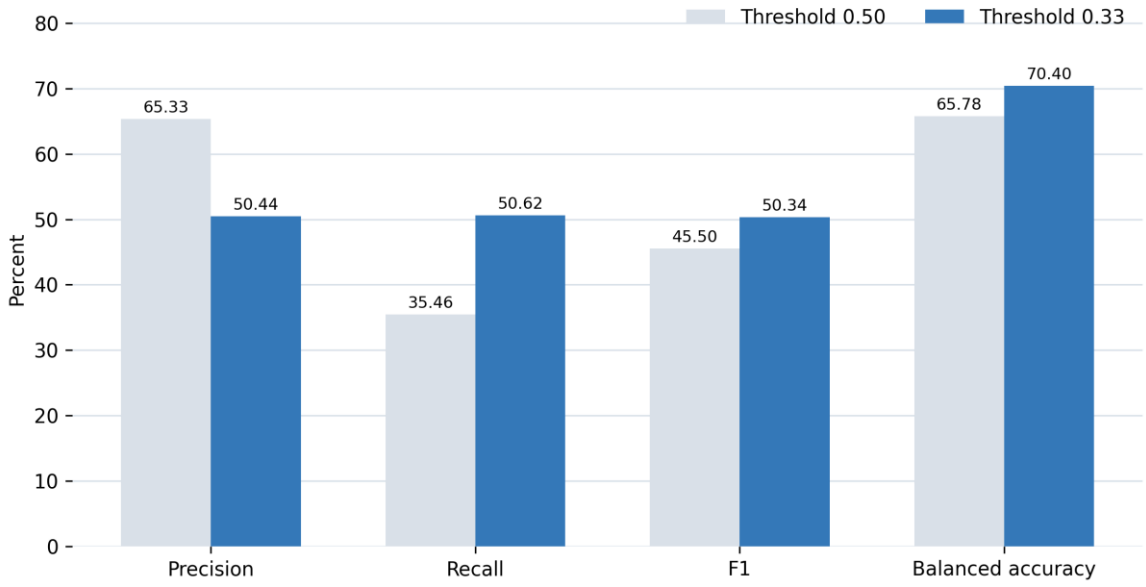


Figure 6.3. Effect of Decision Threshold on CatBoost Metrics

The operational interpretation is also separated into three concepts: probability is the continuous model output in the range [0,1]; predictedAttrition is the thresholded Boolean decision; and riskScore is the percentage-scaled value shown to the recruiter. This separation allows the interface to expose both the raw probability and the selected decision threshold.

The explainability design has two levels. First, the CatBoost service uses native TreeSHAP to calculate local feature contributions for a given prediction. The top five global mean absolute SHAP features in the ablation experiment were OverTime, StockOptionLevel, AverageSatisfaction, NumCompaniesWorked, and DistanceFromHome. Second, a Gemini explanation service grounds the model drivers in the candidate CV and job context. The explanation service is constrained not to recalculate scores or invent candidate facts.

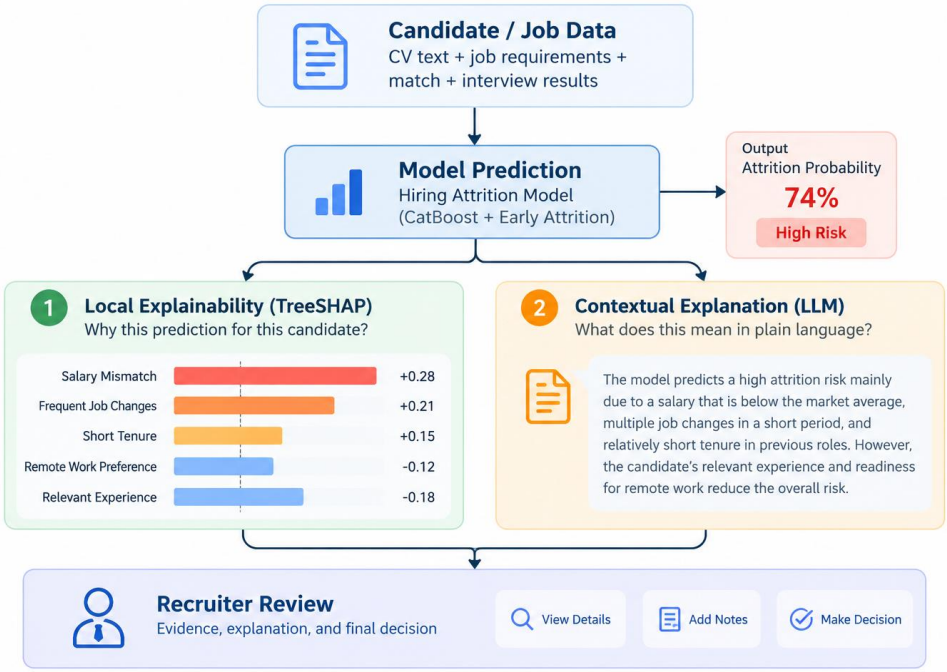


Fig. 8. Two-Level Explainability Workflow.

Figure 6.4. Two-Level Explainability Workflow

This structure is important because a natural-language explanation should not become a second scoring system. The mathematical attribution remains the source of the feature contribution, while the contextual narrative only explains what those model-derived drivers mean in the candidate and job context.

6.2 Product and technical results

The implemented recruiter-facing interface provides separate displays for attrition probability, risk level, threshold status, model agreement, contributing factors, and What-If scenario results. The backend returns model metadata and local drivers, while the explanation layer provides contextual narrative without recalculating the prediction.

6.3 User or stakeholder validation

The existing thesis does not report a separate quantitative user study or stakeholder survey. The available validation is therefore functional and workflow-oriented: the recruiter-facing interface was implemented and exercised through prediction, explanation, and What-If scenarios. A controlled recruiter pilot is identified as a future commercialization and validation step.

6.4 Integration results

The integration workflow connects the recruiter interface, API Gateway, Attrition Orchestrator, CatBoost service, Early Attrition service, explainability service, and supporting candidate/job context. The reported test cases show that the services can share a stable model-ready contract, execute model inference, generate explanations, and re-run the same path for What-If scenarios without mutating the stored candidate snapshot.

6.5 Summary of findings

- Attrition prediction should be separated from candidate suitability rather than used as a direct replacement for the match score.
- Threshold selection materially changes recall in an imbalanced attrition problem, making threshold-aware evaluation more informative than accuracy alone.
- The empirical Sri Lankan survey model provides direct local evidence, while the 1,470-record benchmark supports broader model benchmarking and feature-ablation experiments.
- Feature ablation indicates that adding regional features gave a small, non-significant F1 improvement, while adding the seven engineered ratios reduced F1 significantly in the reported CatBoost experiment.
- Interactive What-If simulation converts the prediction from a passive risk indicator into an exploratory decision-support tool.
- Input coverage and imputation metadata improve transparency when candidate records are incomplete.

Taken together, the reported results show that the attrition component is implementable as a modular decision-support layer. The benchmark supports threshold-aware comparison under class imbalance, the Sri Lankan survey provides local empirical evidence, and the product layer operationalizes prediction with explanation and scenario exploration.

7 DISCUSSION

7.1 Interpretation of findings

First, the technical distinction is that the benchmark model in the production CatBoost case study uses a more extensive feature space and is used for benchmarking models, threshold analysis, and reproducibility purposes. On the other hand, the 835-response Sri Lankan survey model provides stronger local data evidence since it uses records collected from the project questionnaire and includes Sri Lankan employment districts, LKR-based salary variables, and locally designed workplace questions. Separation of these two lines helps in interpreting the evidence.

Secondly, the class distribution affects interpretation of the model. A good overall accuracy cannot serve as evidence of a useful attrition screening when positive cases are relatively rare. As seen in the experiment with threshold, decreasing the threshold value of CatBoost model from 0.50 to 0.33 increased the recall of positive cases from 35.46% to 50.62% and F1 from 45.50% to 50.34%. Hence, the system makes it clear about probability and threshold used for predictions.

Feature ablation showed that additional engineering ratios cannot be considered as helpful to improve model simply because of domain relevance. Adding the seven ratios to the raw and regional configuration in the reported CatBoost experiment decreased F1 score by 3.85 percentage points, the difference being statistically significant. The importance of the fact is that the feature engineering process was treated as an empirical hypothesis.

From the perspective of the decision support system, the value of the system is derived from the combination of prediction, evidence, and scenario exploration. The recruiter can analyze the model drivers, explore the context of the candidate and test different offer conditions without mutating the stored candidate record. In this way, the workflow is more transparent compared to the risk badge only approach while avoiding the assumption that a simulated change will affect retention in reality.

7.2 Comparison with literature baselines and existing solutions

The experimental results also demonstrate that model selection cannot be reduced to the highest accuracy. The production CatBoost model has a useful balance for its intended operational setting, but the Logistic Regression survey model achieved higher ROC-AUC on the local holdout. This distinction should be preserved in future research rather than presenting one model as best on every metric.

The empirical model line and the adapted benchmark line serve different research purposes. The benchmark allows controlled comparison and feature ablation, while the empirical survey provides local evidence. Maintaining both lines prevents the

common methodological mistake of equating synthetic augmentation with real-world validation.

7.3 Research contribution

The research contribution is the development and evaluation of an attrition-risk prediction and decision-support component that separates benchmark and Sri Lankan empirical data lines, applies threshold-aware evaluation, and investigates feature engineering through ablation rather than assuming that additional ratios improve performance.

The component also contributes an operational What-If mechanism and a two-level explanation design in which TreeSHAP/model-derived drivers remain the quantitative evidence while the contextual explanation service translates those drivers into recruiter-readable language.

7.4 Product contribution

The product contribution is the integration of prediction, explanation, scenario simulation, and recruiter review within the wider talent-acquisition workflow. The component adds retention risk as an evidence dimension without replacing the existing candidate matching function and without automatically making a hiring decision.

7.5 Limitations and threats to validity

- The benchmark dataset is adapted/synthetic and originates from the IBM HR Analytics attrition benchmark; therefore its results are not equivalent to real-world Sri Lankan employee-outcome validation.
- The empirical Sri Lankan survey contains 835 responses and 129 positive turnover cases, which limits statistical power and external generalization.
- Candidate data can be incomplete, requiring imputation; the system exposes this through inputCoverage and imputedFeatures, but the quality of a prediction still depends on the available evidence.
- The What-If engine models only the intervention variables implemented in the adapter and does not guarantee real-world behavioural outcomes.
- The current repository audit verifies the prediction, explainability, and simulation services, but a large-scale prospective employer validation study is still required before operational deployment claims can be generalized.

7.6 Implications

In terms of future operational use, the most critical addition is longitudinal validation. Production organization can collect post-hire outcomes and evaluate the prediction quality in terms of calibration, subgroup fairness, drift monitoring, and

organization-specific threshold tuning. Longitudinal validation will also help in testing the simulator by comparing predicted retention with actual outcomes, not just model outputs.

The key design contribution of this component is the clear separation of competence and retention issues. Candidate matching model determines whether the candidate is eligible for the position, while the attrition component estimates the probability of retention based on the current evidence. Therefore, a competent candidate will not be penalized simply due to potential retention issues.

The What-If mechanism is very helpful since many factors that represent risks in the model can be changed by recruiters. Salary modification, work-model options, career development opportunities, or changes in management/mentoring practices can be modeled as counterfactual scenarios. The result will not be the guarantee that intervention prevents the turnover, but a new model inference for the specified scenario.

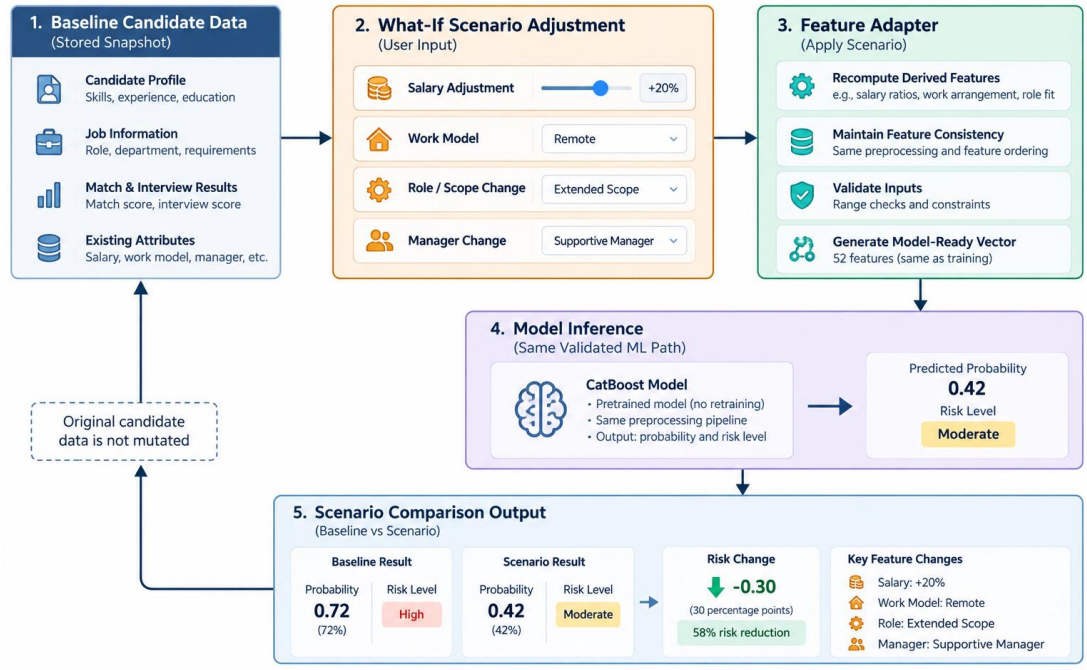


Fig. 9. Interactive What-If Simulation Loop.

Figure 7.1. Interactive What-If Simulation Loop

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

Commercialization can be approached as a recruiter-facing Software as a Service platform in which organizations subscribe to vacancy management, AI screening, attrition decision support, analytics, and reporting features. The individual attrition component is suitable as a value-added risk and retention planning capability that complements, rather than replaces, an organization's existing ATS workflow.

The Hiring Attrition component is suitable for commercialization as an extension of an enterprise talent-acquisition platform. The primary market consists of organizations that need to manage large recruitment pipelines and want to identify retention risks before completing an offer. The component can be positioned as an analytics and decision-support capability rather than an autonomous hiring product.

The proposed value proposition combines prediction, explanation, and scenario planning. Recruiters can inspect the probability and contributing factors, compare the signal with job fit and interview performance, and test selected changes such as salary or working model before finalizing an offer. The business value therefore extends beyond forecasting to structured retention planning.

- Corporate HR departments
- Recruitment agencies
- Talent acquisition teams
- IT companies and startups
- Large enterprises with high-volume recruitment

8.2 Market and competitive position

Software-as-a-Service is a practical primary delivery model because model services, explanation services, and dashboards can be deployed as containers and scaled according to organizational demand. Enterprise deployments can use dedicated environments and role-based administration. An API option can support organizations that already maintain an ATS and want to add a retention-risk service without replacing their existing recruitment system.

- Separate competency matching and attrition-risk analysis
- Explainable model outputs with local feature attribution
- Interactive What-If intervention planning
- Sri Lankan workforce model line for local evidence

8.3 Business or sustainability model

The proposed business model is primarily Software-as-a-Service, complemented by enterprise licensing and API-based decision-support services. The containerized architecture supports modular deployment and scaling according to organizational demand. Sustainability of the analytical service depends on maintaining validated data pipelines, monitoring model performance, and retraining only when sufficient real-world outcome data is available.

8.4 Scalability intellectual property and compliance

The modular architecture allows the model, orchestration, and interface layers to be scaled or replaced independently. Enterprise deployments may use dedicated environments and role-based administration, while organizations with an existing ATS can consume the decision-support capability through an API. The thesis does not claim a registered intellectual-property asset or a compliance certification; instead, it identifies fairness, calibration, data drift, human oversight, and responsible use as requirements for future operational deployment.

8.5 Contribution to the Sustainable Development Goals

Potential alignment can be identified with Sustainable Development Goal 8 (Decent Work and Economic Growth) because the component is intended to support more informed recruitment and retention-oriented decision making. This is an application-level interpretation rather than a measured SDG impact claim. Any future impact assessment should be based on validated organizational outcomes and stakeholder evidence.

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This report presented the Hiring Attrition Prediction and Decision Support component of the AI-Powered Intelligent Talent Acquisition System. The implemented component separates retention risk from candidate suitability and provides an end-to-end workflow spanning candidate-data adaptation, machine-learning inference, local explainability, contextual explanation, and interactive What-If simulation.

The production CatBoost model was benchmarked under class imbalance and tuned to a 0.33 threshold, while separate model tracks were evaluated using the empirical Sri Lankan workforce survey and an Early Attrition classifier. The reported experiments show that threshold-aware evaluation can improve recall, and that the local Sri Lankan survey model provides useful empirical evidence for turnover-related prediction. The What-If mechanism extends the system beyond passive prediction by allowing recruiters to explore how defined changes in salary, work model, role scope, or management conditions alter the model output.

The most important outcome is therefore a transparent decision-support workflow rather than an automatic hiring rule. The system presents prediction, explanation, and recommendations while keeping the recruiter in control of the final decision. Future work should prioritize longitudinal validation with verified organizational outcomes, stronger calibration analysis, broader Sri Lankan datasets, and prospective evaluation with real HR users.

9.2 Achievement of objectives

The objectives were addressed through the preparation and analysis of an empirical Sri Lankan survey and an adapted benchmark dataset, construction of a candidate-to-model feature adapter, engineering and evaluation of domain interaction features, benchmarking of multiple algorithms, threshold selection, What-If simulation, explainability, backend orchestration, and recruiter-facing interface integration. The reported functional tests further verify the main system behavior.

9.3 Individual research and product contributions

The individual contribution covered research formulation, dataset processing, feature engineering, model development and benchmarking, threshold selection, backend orchestration, explainability, What-If simulation, testing, and integration of the recruiter interface.

The combined research and product contribution is an end-to-end attrition decision-support layer that connects model development with recruiter-facing evidence while preserving human responsibility for the final employment decision.

9.4 Future directions

- Collect longitudinal employer outcome data and conduct prospective validation.
- Evaluate probability calibration and organization-specific threshold tuning.
- Assess subgroup fairness and monitor model performance for data drift.
- Expand Sri Lankan datasets with verified recruitment and employment outcomes.
- Evaluate usability, latency, and recruiter trust through controlled stakeholder studies.
- Validate the What-If simulator against observed outcomes rather than model outputs alone.

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GLOSSARY

Attrition. Employee or candidate departure risk considered in the decision-support context.

Decision threshold. A probability cutoff used to convert a continuous model probability into a binary predicted outcome.

Feature Adapter. The processing layer that maps candidate and job information into a stable model-ready representation while handling missing values and scenario overrides.

Holdout set. An untouched evaluation subset used to estimate model performance after model-selection work.

PR-AUC. Area under the precision-recall curve, useful for evaluating ranking performance when the positive class is relatively rare.

TreeSHAP. A model-attribution method used to estimate local feature contributions for tree-based model predictions.

What-If simulation. An interactive scenario analysis in which selected inputs are changed and the prediction is recalculated without changing the stored candidate snapshot.

APPENDICES

Appendix A Research instruments or extended results

The thesis uses two research data lines described in Chapter 3: an empirical Sri Lankan workforce survey of 835 responses and an adapted/synthetic benchmark of 1,470 records. The extended functional verification table is presented in Chapter 5, while detailed benchmark and holdout metrics are presented in Chapter 6. The figures and tables in the main thesis provide the principal evidence required to interpret the reported experiments.

Appendix B Academic integrity and AI use disclosure

Generative AI and other automated tools were used for limited academic-support activities including language refinement, document restructuring, and formatting assistance in preparing this thesis according to the supplied SLIIT template. A representative workflow was to reorganize the existing individual thesis content into the required template sections without introducing new empirical results. The author remains responsible for the research design, data, code, citations, analyses, interpretations, and conclusions, and should verify all final content before submission.